

A Case Study of Analyzing Wind Data Proposing a Wind Farm Design to Provide Zero-Carbon Energy to Hettinger City.

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Abstract

Analyzing wind data of Hettinger City, North Dakota, to design a wind farm that can utilize renewable energy to generate electricity, thus increasing the sustainability of the area. This report accounts for environmental impact, licenses, zoning regulations, installation costs and capacity factor in the process of the design, thus providing the complete outline involving the design process.

Introduction

The area chosen for the analysis and design of our wind farm was Hettinger City in North Dakota seen in Figure 1. According to the United States Census Bureau, the population is roughly 1,051 as of July 2021. The state of North Dakota ranks top 10 in average wind speeds which makes it a great place for wind farms that can power entire cities and even potentially provide extra power back to the grid. This report shows the production for 100% of the electricity load of Hettinger which is roughly 32 GWh.



Figure 1: Hettinger City, ND

Electrical Energy Demand

According to Find Energy, the total energy consumption in Hettinger was 32.830 GWh as of 2021. With a population of 1050, this consumption is split between industrial, commercial and residential. Demand does not remain constant through the year due to seasonal changes in electricity demand. Seen below in Figure 2 is a monthly energy estimation of Hettinger. As seen in the chart, energy consumption is higher during the winter due to increased heating demands.

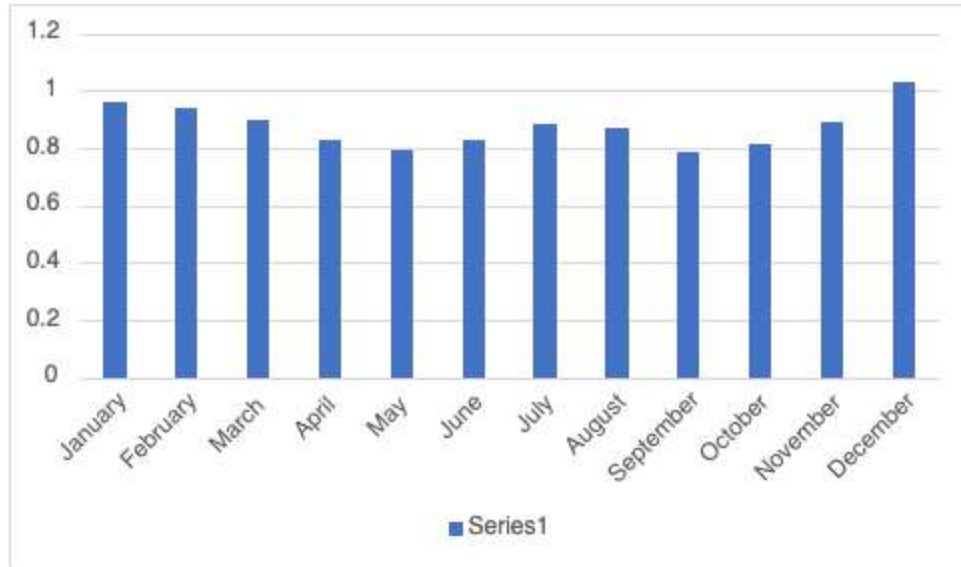


Figure 2: Estimated Monthly Energy Demand

Wind Data Analysis

The raw 50M and shear exponent calculations for 100M are provided in the reference to our raw data. Below is the shear exponent formula used in this calculation.

$$u_{100} = u_{50} \left(\frac{100}{50} \right)^{\left(\frac{1}{7} \right)}$$

This can be used to create a data bin that gives the frequency of a certain wind speed as seen in Figure 3. This is then compared to a Rayleigh's distribution to examine the similarities in the probability functions used.

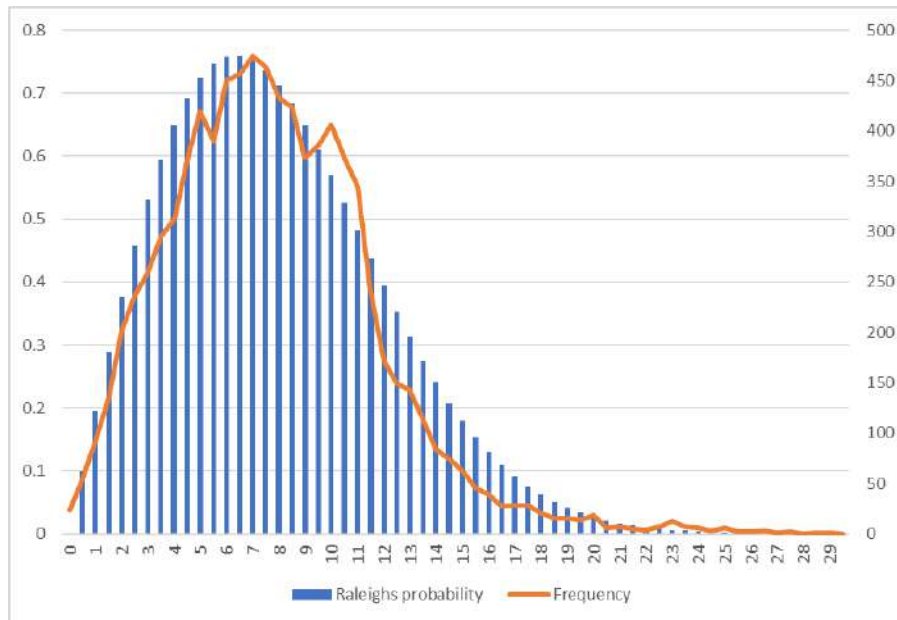


Figure 3: Wind Speed vs Frequency & Rayleigh's Probability

Month	Monthly Averages	SD Monthly	TI Monthly
January	8.724117627	4.986775221	0.571607976
February	8.271125579	3.783380934	0.457420323
March	8.522650971	4.395417916	0.515733653
April	7.843865939	3.78915406	0.483072262
May	7.523197683	3.357934335	0.446344025
June	7.301589307	3.84789964	0.526994806
July	6.802734755	2.89340125	0.425329129
August	7.413263608	3.254485687	0.439008493
September	7.658624254	2.939639027	0.383833823
October	8.727812766	3.604253226	0.412961795
November	9.520103839	4.70460034	0.494175318
December	8.099966702	4.008365443	0.494861965

Table 1: Wind Speed Calculations for 100M

Annual Average	SD yearly	TI yearly
8.034087752	3.911488303	0.486861536

Table 2: Annual Average, Standard Deviation, Turbulence Intensity Calculations for 100M

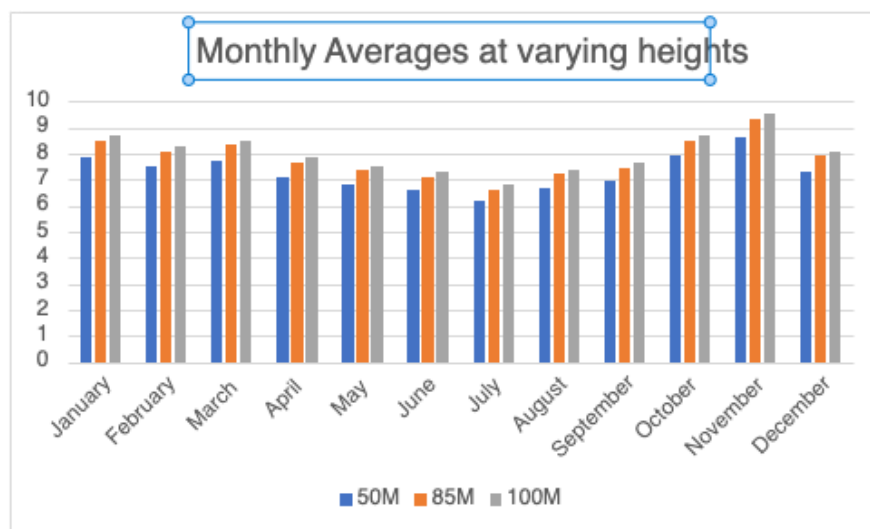


Figure 4: Monthly Averages for 50, 85 and 100M

Turbine Selection

The turbine that provided a capacity factor of 42% at height of 100M was the Vestas turbine (V 100/1800), rated at 1800 kW with a rotor diameter of 100M. The below power curve is for V100/1800, with a polynomial estimation in Figure 6.

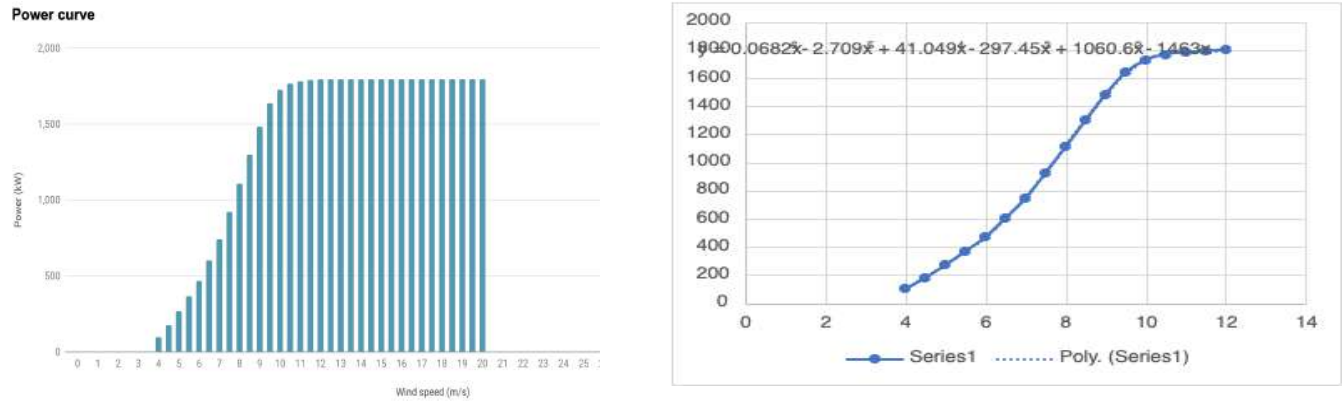


Figure 5 & 6: Power curve of V100/1.8 and Polynomial approximation of the 6th order

Environmental Impact

It is standard nowadays to study the impact any new construction project will have on its surrounding environment, along with following necessary regulations and permits. This should be priority number one when choosing a site for a wind farm. Wildlife disturbances, historical or culturally sensitive sites, etc are all issues that need to be researched when picking a site for a new wind farm.

According to the State Historical Society of North Dakota, there are no historical sites within a 50 mile radius of Hettinger, ND. We also took into consideration that North Dakota is home to five Native American tribes when choosing our wind farm location. Our site also does not fall within any Indian reservations or culturally sensitive sites. Since the area is rural farm land and because of the points mentioned above, impact to viewshed will be minimal.

Because of the importance and value of renewable energies, there are many government sponsored studies that look at environmental impacts of all forms in renewable energies, including wind. Tethys, developed by the Pacific Northwest National Laboratory, has done most of the heavy lifting for us already in terms of environmental impact research. With North Dakota, the main species of animals to take into account are the Whooping crane and waterfowl. Waterfowl breeding locations are also crucial habitats for many wetland-dependent birds in North America, so the majority of predicted breeding areas were excluded as usable land in Figure X below. Tether also excluded whooping crane stopover sites as usable land, along with a

1.6k km buffer area around the Missouri and Red Rivers which are important migratory corridors for birds and bats. Thus, the low impact area graph was created in Figure 7 which shows that Hettinger, represented by the red dot, is in a low impact area.

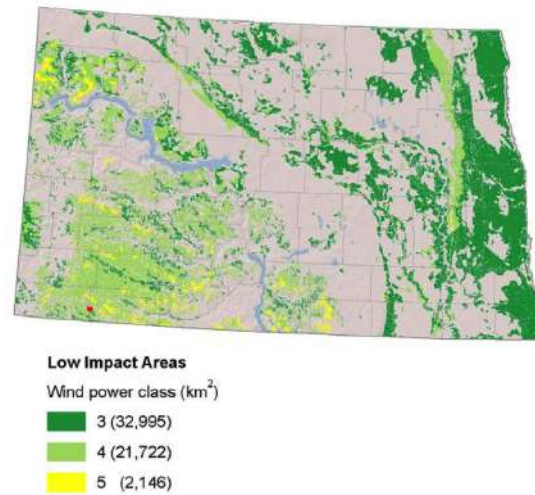


Figure 7: Low impact areas for wind development in North Dakota.

Permits

According to Hettinger's zoning ordinance, our proposed location falls within the agriculture district. Within the agriculture district, a permitted use of land is for alternative energy production devices, such as wind turbines. We plan to pay the landowner \$40,000 a year to rent the land. Because of this supplemental income we'll be offering to the owner, we anticipate being able to partner with the landowner with minimal pushback. The landowner will also still be able to use the land for agriculture purposes.

Some additional requirements our find warm would have to follow are as follows:

- The minimum lot area with an Agricultural district shall be 5 acres
- Minimum front, rear and side yard setback shall be 50 feet from the property

Wind Farm Expenses

Turbine Distribution & Direction

To provide maximum power output by reducing discrepancies caused by wake effect. To get optimal energy production, the turbines are placed, 3.5 times the rotor diameter (r_d), perpendicular to wind direction and 7 times the r_d in the direction of the wind. With our r_d as 100m, the placement is as seen in Figure 8. This allows for a wake effect at 9.3 % for the back row.

$$\begin{aligned}
 \text{Total Energy generation} &= 3 * 6780 + (1 - X_{Wake}) * 2 * 6780 \\
 &= 32800 \text{ MWh}
 \end{aligned}$$

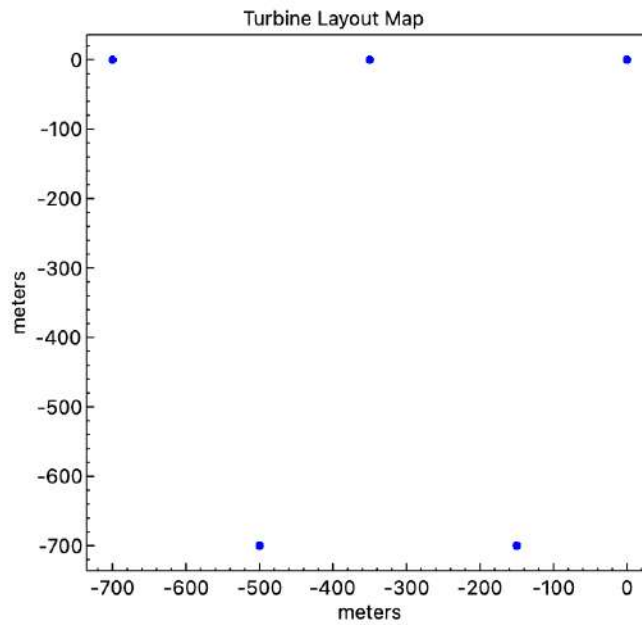


Figure 8: Turbine placement



Figure 9: Topographic representation

The satellite image shows the area covered by the turbine but we chose to buy 121 acres of land which is close to 400,000 meter squared. This is also only 500m away from a substation.

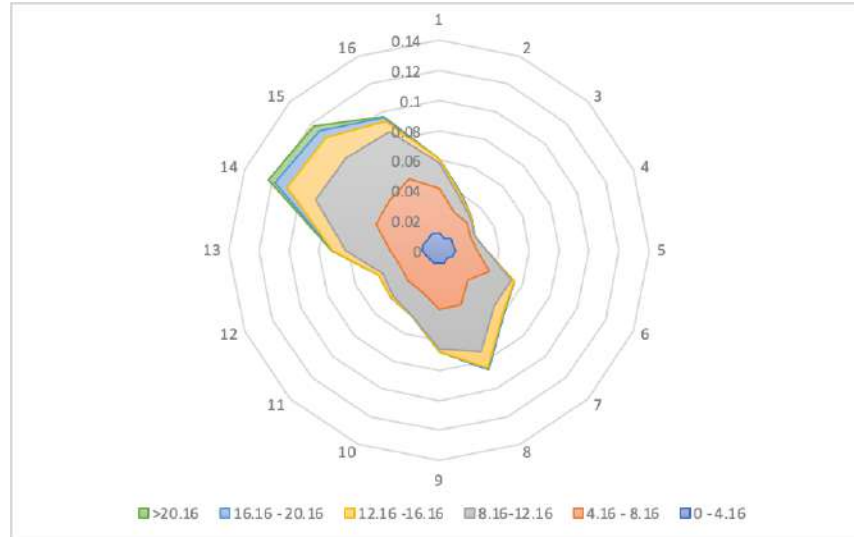


Figure 10: Wind Rose for turbine direction placement

The wind rose shows the direction of the wind speeds, with greater speeds coming mostly from the North-West. This helps to angle the wind turbine in such a direction to have greater efficiency.

Installation Costs

The installations costs are for the V100/1.8, with a 1800 kW rated power, 100M hub height, 100M r_d and a capacity factor of 43%. The below pie chart is a total representation of the CapEx or Capital Expenditure for the turbine. This would be scaled to show the estimate of 5 turbines in the wind farm.

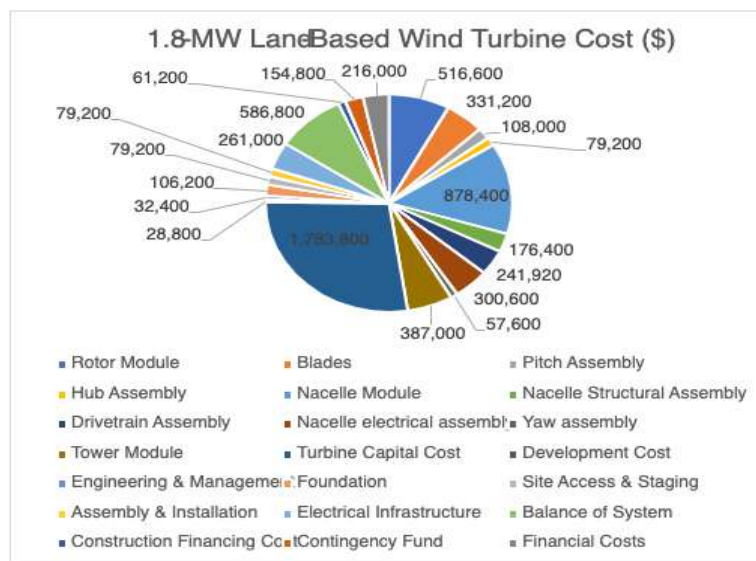


Figure 11: Total Installation costs for 1.8 MW turbine

O&M Expenses

The average wind project has an O&M cost of \$0.01 to \$0.02 per kWh produced. A wind turbine has an energy production of 6780 MWh/yr and with \$0.01 per kWh, the cost per turbine is \$77,400 per year.

Levelized Cost of Energy (LCOE)

The LCOE or the Levelized Cost of Energy is a measure of the cost of energy produced. It helps in the comparison between energy systems.

$$LCOE = \frac{(CapEx * FCR) + OpEx}{AEP_{net}/1000}$$

$$LCOE = \frac{(2,585,000 * 6.5\%) + 77400}{3766.7/1000}$$

$$LCOE = 33.80 \$/MWh$$

Losses

There are various losses as seen below that account to 15%. Losses include wake, availability, electricity, turbine performance, environmental and curtailment.

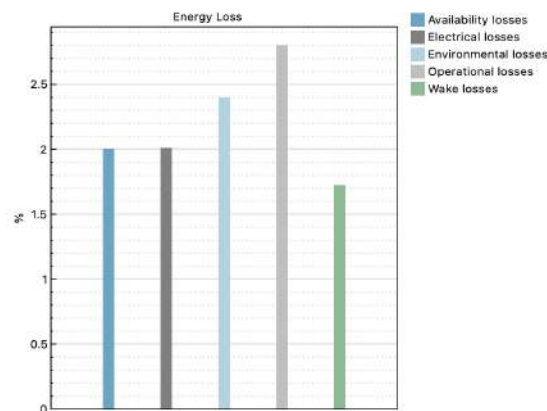


Figure 12: Losses

Tax Expenses

We calculated an expected property tax of \$57,600 per year. This is calculated using the given value of \$6400 per MW nameplate capacity. The state tax is estimated to be \$164,000 per year in the state of North Dakota with a \$6400 per MW for 5 turbines.

To help offset some of these taxes, there is a federal tax credit. As of Dec. 31, 2021, the federal tax credit is 2.5 cents/KWh. In total this tax credit would give us \$847,500 per year. For sake of simplicity and inability to predict the future, we are planning for the tax credit amount per kWh to remain the same throughout our 10 years.

Land Leasing and Tax

The required area of land is 121 acres or $490,000 \text{ m}^2$. With a \$12,500 per acre, the lease per year costs \$1,512,500. The land leasing tax is approximately \$8000 per turbine and a total of \$40,000 for 5 turbines.

Expected Energy Production

The Vestas turbine V100/1.8 selected at given wind speeds produces the energy as seen below in kWh.

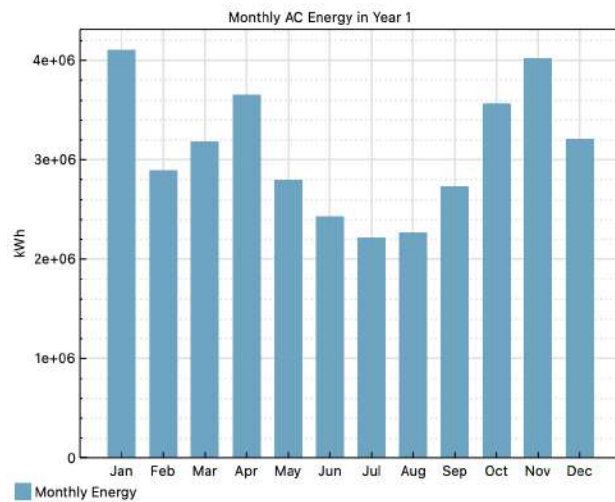


Figure 13: Monthly Energy Production

Break-Even Analysis

CapEx: \$12,920,000

Land Leasing: 121 acres : \$12,500/acre = \$1,512,500/yr

Property Tax = \$6400/MW * 9MW = \$57,600/yr

Land Lease Tax = \$8000/turbine * 5 turbines = \$40,000/yr

State Tax = \$0.05/MW * 328000 = \$164,000/yr

$$\text{Total Cost} = \$1,512,500 + \$57,600 + \$40,000 + \$164,000 = \$1,744,100/\text{yr}$$

$$A = \frac{(1+4\%)^{10} - 1}{4\%(1+4\%)^{10}} = 8.1109$$

$$P = A(P/A, 4\%, 10) = \$1,744,100/\text{yr} * 8.1109 = \$14,389,723$$

$$\text{Total} = \$14,389,723 + \$12,920,000 = \$27,309,723$$

$$\text{Retail (cent/kWh)} = \frac{\$27,309,723}{32,800 * 10^3 * 8.1109} = 10.265 \text{ cent/kWh}$$

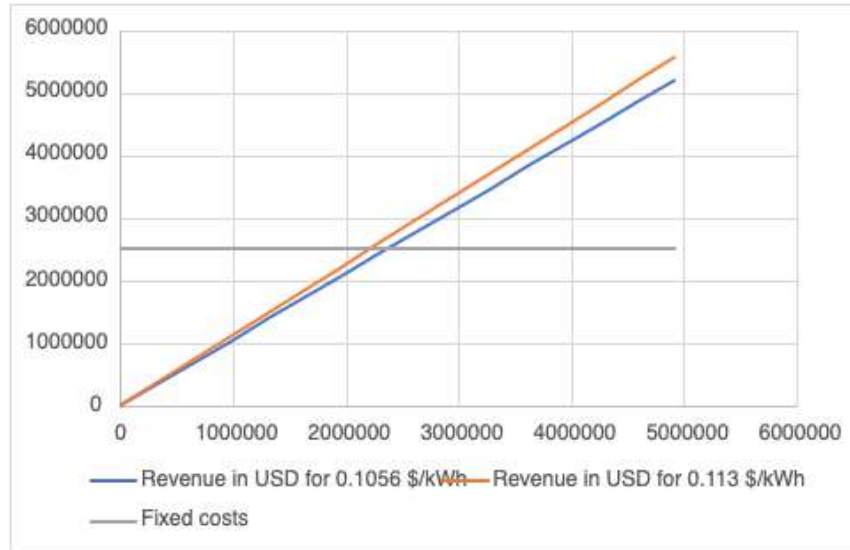


Figure 14: kWh vs Revenue earned for 10.2 cents and 11.3 cents

Thus to reach the break even point at 10 years, we would have to sell at 10.265 cents/kWh. This is 9% less than the national average which is 11.3 cents/kWh. The above breakeven is not in reference to time but the revenue at which the break in occurs for 10.265 cents and 11.3 cents.

Summary

Our wind farm is estimated to produce 32.8 GWh annually which will be sold at 10.265 cents/kWh. This is below the national average that Americans currently pay for electricity. On top of our power being cheaper, the other benefit is that this is 100% clean, renewable energy being generated for the town of Hettinger. Our annual revenue is estimated to be \$3,366,920 along with a \$847,500 tax credit we'll receive each year. With a sum of yearly expenses calculated out to be \$27,309,723, we anticipate to break even at the 10 year mark as long as we sell our electricity at 10.265 cents/kWh. With most wind farms lasting between 20-25 years, not only do we see this as a good investment for the town of Hettinger, but we also see it as a good personal investment.

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Excel Sheet Calculations: [Wind_data_Hettinger.xlsx](#)

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